1) a)

$$\int \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x = \arcsin x + c$$

1) b)

$$\int \frac{1}{1+x^2}\,\mathrm{d}x = \arctan x + c$$

1) c)

$$\int \frac{1}{\sqrt{x^2-1}}\,\mathrm{d}x = \operatorname{arcosh} x + c$$

1) d)

$$\int \frac{1}{\sqrt{1+x^2}}\,\mathrm{d}x = \operatorname{arsinh} x + c$$

1) e)

$$\int \frac{1}{1-x^2}\,\mathrm{d}x = \operatorname{artanh} x + c$$

2) a)

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 6\cot x\,\mathrm{d}x$$

$$= 6\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos x}{\sin x}\,\mathrm{d}x$$

$$= 6\left[\ln|\sin x|\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= 6\left(\ln\left|\sin\frac{\pi}{2}\right| - \ln\left|\sin\frac{\pi}{4}\right|\right)$$

$$= 6\left(\ln 1 - \ln\frac{\sqrt{2}}{2}\right)$$

$$= -6\ln\frac{\sqrt{2}}{2}$$

$$= \ln\left(\left(\frac{\sqrt{2}}{2}\right)^{-6}\right)$$

$$= \ln 8$$

2) b)

$\int_0^{\frac{\pi}{2}} 6\cot x\,\mathrm{d}x$ is an improper integral because $\cot 0$ is undefined.

$\therefore \int_0^{\frac{\pi}{2}} 6\cot x\,\mathrm{d}x$

$= \lim_{a\to 0} \int_a^{\frac{\pi}{2}} 6\cot x\,\mathrm{d}x$

$= \lim_{a\to 0} [6\ln|\sin x|]_a^{\frac{\pi}{2}}$

$= \lim_{a\to 0} \left(6\ln\left|\sin\frac{\pi}{2}\right| - 6\ln|\sin a|\right)$

As $a \to 0$, $\sin a \to 0$, so $\ln|\sin a| \to -\infty$

So the integral diverges, and so $\int_0^{\frac{\pi}{2}} 6\cot x\,\mathrm{d}x$ is undefined.

3) a)

Use Integration by Substitution:

$u = \tan x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \sec^2 x \Rightarrow \mathrm{d}x = \cos^2 x\,\mathrm{d}u$

$\therefore \int \frac{1}{25\cos^2 x + \sin^2 x}\,\mathrm{d}x$

$= \int \frac{\frac{1}{\cos^2 x}}{25 + \frac{\sin^2 x}{\cos^2 x}}\,\mathrm{d}x$

$= \int \frac{\frac{1}{\cos^2 x}}{25 + \tan^2 x}\,\mathrm{d}x$

$= \int \frac{\frac{1}{\cos^2 x}}{25 + u^2} \times \cos^2 x\,\mathrm{d}u$

$= \int \frac{1}{25+u^2}\,\mathrm{d}u$

3) b)

Change limits:

$x = \arctan 5 \Rightarrow u = \tan(\arctan 5) = 5$

$x = 0 \Rightarrow u = \tan 0 = 0$

$\therefore \int_0^{\arctan 5} \frac{1}{25\cos^2 x + \sin^2 x}\,\mathrm{d}x$

$= \int_0^5 \frac{1}{25+u^2}\,\mathrm{d}u$

$= \int_0^5 \frac{1}{25\left(1+\frac{1}{25}u^2\right)}\,\mathrm{d}u$

$= \frac{1}{25}\int_0^5 \frac{1}{1+\left(\frac{1}{5}u\right)^2}\,\mathrm{d}u$

$= \frac{1}{25}\left[5\arctan\left(\frac{1}{5}u\right)\right]_0^5$

$= \frac{1}{25}\left(5\arctan\left(\frac{1}{5}\times 5\right) - 5\arctan\left(\frac{1}{5}\times 0\right)\right)$

$= \frac{1}{25}(5\arctan 1 - 0)$

$= \frac{1}{25}\times\frac{5\pi}{4}$

$= \frac{\pi}{20}$

So $k = \frac{1}{20}$

4)

$I = \int_{\sqrt{\ln 2}}^{\sqrt{\ln 5}} \frac{x}{\sinh(x^2)}\,\mathrm{d}x$

Use Integration by Substitution:

Let $u = x^2 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

Change Limits:

$x = \sqrt{\ln 5} \Rightarrow u = \ln 5$

$x = \sqrt{\ln 2} \Rightarrow u = \ln 2$

$\therefore I = \int_{\ln 2}^{\ln 5} \frac{x}{\sinh u}\times\frac{1}{2x}\,\mathrm{d}u$

$= \frac{1}{2}\int_{\ln 2}^{\ln 5} \frac{1}{\sinh u}\,\mathrm{d}u$

$= \frac{1}{2}\int_{\ln 2}^{\ln 5} \frac{2}{\mathrm{e}^u - \mathrm{e}^{-u}}\,\mathrm{d}u$

$= \frac{1}{2}\int_{\ln 2}^{\ln 5} \frac{2\mathrm{e}^u}{\mathrm{e}^{2u}-1}\,\mathrm{d}u$

Use Integration by Substitution:

Let $v = \mathrm{e}^u \Rightarrow \frac{\mathrm{d}v}{\mathrm{d}u} = \mathrm{e}^u \Rightarrow \mathrm{d}u = \frac{1}{\mathrm{e}^u}\mathrm{d}v = \frac{1}{v}\mathrm{d}v$

Change Limits:

$u = \ln 5 \Rightarrow v = 5$

$u = \ln 2 \Rightarrow v = 2$

$\therefore I = \frac{1}{2}\int_2^5 \frac{2v}{v^2-1} \times \frac{1}{v}\,\mathrm{d}v$

$= \int_2^5 \frac{1}{v^2-1}\,\mathrm{d}v$

Note that the limits of the integral are outside the domain of $\operatorname{artanh} v$, so we must instead use partial fractions:

$\frac{1}{v^2-1} \equiv \frac{1}{(v-1)(v+1)} \equiv \frac{A}{v-1} + \frac{B}{v+1}$

$\Rightarrow 1 \equiv A(v+1) + B(v-1)$

Let $v = 1$: $1 = 2A \Rightarrow A = \frac{1}{2}$

Let $v = -1$: $1 = -2B \Rightarrow B = -\frac{1}{2}$

$\therefore \frac{1}{v^2-1} \equiv \frac{\frac{1}{2}}{v-1} - \frac{\frac{1}{2}}{v+1}$

So:

$I = \frac{1}{2}\int_2^5 \left(\frac{1}{v-1} - \frac{1}{v+1}\right)\mathrm{d}v$

$= \frac{1}{2}\left[\ln|v-1| - \ln|v+1|\right]_2^5$

$= \frac{1}{2}\left((\ln 4 - \ln 6) - (\ln 1 - \ln 3)\right)$

$= \frac{1}{2}\left(\ln\frac{2}{3} + \ln 3\right)$

$= \frac{1}{2}\ln 2$

5) a)

$x^2 - 12x + 61$

$\equiv (x-6)^2 - 36 + 61$

$\equiv (x-6)^2 + 25$

$\int \frac{1}{x^2-12x+61}\,\mathrm{d}x$

$= \int \frac{1}{25+(x-6)^2}\,\mathrm{d}x$

$= \int \frac{1}{25\left(1+\frac{1}{25}(x-6)^2\right)}\,\mathrm{d}x$

$= \frac{1}{25}\int \frac{1}{1+\left(\frac{1}{5}x-\frac{6}{5}\right)^2}\,\mathrm{d}x$

$= \frac{1}{25} \times \frac{1}{\frac{1}{5}} \times \arctan\left(\frac{1}{5}x - \frac{6}{5}\right) + c$

$= \frac{1}{5}\arctan\left(\frac{1}{5}x - \frac{6}{5}\right) + c$

5) b)

$\int_5^7 \left(3 - \frac{78}{x^2-12x+61}\right)\mathrm{d}x$

$= \left[3x - \frac{78}{5}\arctan\left(\frac{1}{5}x - \frac{6}{5}\right)\right]_5^7$

$= \left(3\times 7 - \frac{78}{5}\arctan\frac{1}{5}\right) - \left(3\times 5 - \frac{78}{5}\arctan\left(-\frac{1}{5}\right)\right)$

$= \left(21 - \frac{78}{5}\arctan\frac{1}{5}\right) - \left(15 + \frac{78}{5}\arctan\frac{1}{5}\right)$

$= 6 - \frac{156}{5}\arctan\frac{1}{5}$

5) c) i)

$0 = 3 - \frac{78}{x^2-12x+61}$

$\Rightarrow 0 = 1 - \frac{26}{x^2-12x+61}$

$\Rightarrow 0 = x^2 - 12x + 61 - 26$

$\Rightarrow 0 = x^2 - 12x + 35$

$\Rightarrow 0 = (x-5)(x-7)$

$\Rightarrow x = 5$ or $x = 7$

So $P(5,0)$ and $Q(7,0)$

5) c) ii)

$y = 3 - \frac{78}{x^2-12x+61}$

$\Rightarrow y = 3 - 78(x^2 - 12x + 61)^{-1}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -78 \times (-1) \times (2x - 12) \times (x^2 - 12x + 61)^{-2}$

$= \frac{156(x-6)}{(x^2-12x+61)^2}$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\frac{156(x-6)}{(x^2-12x+61)^2} = 0$

$\Rightarrow x = 6$

$y|_{x=6} = 3 - \frac{78}{6^2-12\times6+61} = -\frac{3}{25}$

So $R\left(6, -\frac{3}{25}\right)$

5) c) iii)

$\int_5^6 \left(3 - \frac{78}{x^2-12x+61}\right) \mathrm{d}x$

$= \left[3x - \frac{78}{5}\arctan\left(\frac{1}{5}x - \frac{6}{5}\right)\right]_5^6$

$= \left(3 \times 6 - \frac{78}{5}\arctan 0\right) - \left(3 \times 5 - \frac{78}{5}\arctan\left(-\frac{1}{5}\right)\right)$

$= (18 - 0) - \left(15 + \frac{78}{5}\arctan\frac{1}{5}\right)$

$= 3 - \frac{78}{5}\arctan\frac{1}{5}$

However, as the region is below the x-axis, the exact area is

$\left(\frac{78}{5}\arctan\frac{1}{5} - 3\right)$ units^2

6) a)

$\frac{2x^3+x^2+x}{(x+1)(x^2+1)} \equiv A + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}$

$\Rightarrow 2x^3 + x^2 + x \equiv A(x+1)(x^2+1) + B(x^2+1) + (Cx+D)(x+1)$

Let $x = -1$: $-2 = 2B \Rightarrow B = -1$

Let $x = 0$: $0 = A - 1 + D \Rightarrow A + D = 1$

Let $x = 1$: $4 = 4A - 2 + 2C + 2D \Rightarrow 2A + C + D = 3$

Let $x = 2$: $22 = 15A - 5 + (2C + D) \times 3 \Rightarrow 15A + 6C + 3D = 27$

$A = 2, C = 0, D = -1$

$\therefore \int \frac{2x^3+x^2+x}{(x+1)(x^2+1)} \mathrm{d}x$

$= \int \left(2 - \frac{1}{x+1} - \frac{1}{x^2+1}\right) \mathrm{d}x$

$= 2x - \ln|x+1| - \arctan x + c$

6) b)

$\int_0^1 \frac{2x^3+x^2+x}{(x+1)(x^2+1)} \mathrm{d}x$

$= [2x - \ln|x+1| - \arctan x]_0^1$

$= (2 \times 1 - \ln|1+1| - \arctan 1) - (2 \times 0 - \ln|0+1| - \arctan 0)$

$= \left(2 - \ln 2 - \frac{\pi}{4}\right) - (0 - 0 - 0)$

$= 2 - \ln 2 - \frac{\pi}{4}$

7) a)

$I = \int_{-\frac{5}{2}}^{-\frac{3}{2}} \frac{1}{\sqrt{4x^2+20x+34}}\,\mathrm{d}x$

$4x^2 + 20x + 34$

$\equiv 4(x^2 + 5x) + 34$

$\equiv 4\left(\left(x + \frac{5}{2}\right)^2 - \frac{25}{4}\right) + 34$

$\equiv 4\left(x + \frac{5}{2}\right)^2 - 25 + 34$

$\equiv 4\left(x + \frac{5}{2}\right)^2 + 9$

$\therefore I = \int_{-\frac{5}{2}}^{-\frac{3}{2}} \frac{1}{\sqrt{9+4\left(x+\frac{5}{2}\right)^2}}\,\mathrm{d}x$

$= \int_{-\frac{5}{2}}^{-\frac{3}{2}} \frac{1}{\sqrt{9\left(1+\frac{4}{9}\left(x+\frac{5}{2}\right)^2\right)}}\,\mathrm{d}x$

$= \frac{1}{3}\int_{-\frac{5}{2}}^{-\frac{3}{2}} \frac{1}{\sqrt{1+\left(\frac{2}{3}x+\frac{5}{3}\right)^2}}\,\mathrm{d}x$

Use Integration by Substitution:

Let $\sinh u = \frac{2}{3}x + \frac{5}{3} \Rightarrow \cosh u \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{3} \Rightarrow \mathrm{d}x = \frac{3}{2}\cosh u\,\mathrm{d}u$

Change limits:

$x = -\frac{3}{2} \Rightarrow u = \operatorname{arsinh}\left(\frac{2}{3}\left(-\frac{3}{2}\right) + \frac{5}{3}\right) = \operatorname{arsinh}\frac{2}{3}$

$x = -\frac{5}{2} \Rightarrow u = \operatorname{arsinh}\left(\frac{2}{3}\left(-\frac{5}{2}\right) + \frac{5}{3}\right) = 0$

$\therefore I = \frac{1}{3}\int_0^{\operatorname{arsinh}\frac{2}{3}} \frac{\frac{3}{2}\cosh u}{\sqrt{1+\sinh^2 u}}\,\mathrm{d}u$

$= \frac{1}{2}\int_0^{\operatorname{arsinh}\frac{2}{3}} \frac{\cosh u}{\cosh u}\,\mathrm{d}u$

$= \frac{1}{2}\int_0^{\operatorname{arsinh}\frac{2}{3}} 1\,\mathrm{d}u$

$= \frac{1}{2}[u]_0^{\operatorname{arsinh}\frac{2}{3}}$

$= \frac{1}{2}\left(\operatorname{arsinh}\frac{2}{3} - 0\right)$

$= \frac{1}{2}\left(\ln\left(\frac{2}{3} + \sqrt{\left(\frac{2}{3}\right)^2 + 1}\right) - 0\right)$

$= \frac{1}{2}\ln\left(\frac{2+\sqrt{13}}{3}\right)$

7) b)

$I = \int_{\frac{1}{2}}^{\frac{5}{4}} \frac{1}{\sqrt{-4x^2+4x+8}}\,\mathrm{d}x$

$-4x^2 + 4x + 8$

$\equiv -4(x^2 - x) + 8$

$\equiv -4\left(\left(x - \frac{1}{2}\right)^2 - \frac{1}{4}\right) + 8$

$\equiv -4\left(x - \frac{1}{2}\right)^2 + 1 + 8$

$\equiv -4\left(x - \frac{1}{2}\right)^2 + 9$

$\therefore I = \int_{\frac{1}{2}}^{\frac{5}{4}} \frac{1}{\sqrt{9-4\left(x-\frac{1}{2}\right)^2}}\,\mathrm{d}x$

$= \int_{\frac{1}{2}}^{\frac{5}{4}} \frac{1}{\sqrt{9\left(1-\frac{4}{9}\left(x-\frac{1}{2}\right)^2\right)}}\,\mathrm{d}x$

$= \frac{1}{3}\int_{\frac{1}{2}}^{\frac{5}{4}} \frac{1}{\sqrt{1-\left(\frac{2}{3}x-\frac{1}{3}\right)^2}}\,\mathrm{d}x$

Use Integration by Substitution:

Let $\sin u = \frac{2}{3}x - \frac{1}{3} \Rightarrow \cos u \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{3} \Rightarrow \mathrm{d}x = \frac{3}{2}\cos u\,\mathrm{d}u$

Change limits:

$x = \frac{5}{4} \Rightarrow u = \arcsin\left(\frac{2}{3} \times \frac{5}{4} - \frac{1}{3}\right) = \frac{\pi}{6}$

$x = \frac{1}{2} \Rightarrow u = \arcsin\left(\frac{2}{3} \times \frac{1}{2} - \frac{1}{3}\right) = 0$

$\therefore I = \frac{1}{3}\int_0^{\frac{\pi}{6}} \frac{\frac{3}{2}\cos u}{\sqrt{1-\sin^2 u}}\,\mathrm{d}u$

$= \frac{1}{2}\int_0^{\frac{\pi}{6}} \frac{\cos u}{\cos u}\,\mathrm{d}u$

$= \frac{1}{2}\int_0^{\frac{\pi}{6}} 1\,\mathrm{d}u$

$= \frac{1}{2}[u]_0^{\frac{\pi}{6}}$

$= \frac{1}{2}\left(\frac{\pi}{6} - 0\right)$

$= \frac{\pi}{12}$

7) c)

$I = \int_{\frac{\sqrt{3}-3}{2}}^{0} \frac{1}{4x^2+12x+18}\,\mathrm{d}x$

$4x^2 + 12x + 18$

$\equiv 4(x^2 + 3x) + 18$

$\equiv 4\left(\left(x + \frac{3}{2}\right)^2 - \frac{9}{4}\right) + 18$

$\equiv 4\left(x + \frac{3}{2}\right)^2 - 9 + 18$

$\equiv 4\left(x + \frac{3}{2}\right)^2 + 9$

$\therefore I = \int_{\frac{\sqrt{3}-3}{2}}^{0} \frac{1}{9+4\left(x+\frac{3}{2}\right)^2}\,\mathrm{d}x$

$= \int_{\frac{\sqrt{3}-3}{2}}^{0} \frac{1}{9\left(1+\frac{4}{9}\left(x+\frac{3}{2}\right)^2\right)}\,\mathrm{d}x$

$= \frac{1}{9}\int_{\frac{\sqrt{3}-3}{2}}^{0} \frac{1}{1+\left(\frac{2}{3}x+1\right)^2}\,\mathrm{d}x$

Use Integration by Substitution:

Let $\tan u = \frac{2}{3}x + 1 \Rightarrow \sec^2 u \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{3} \Rightarrow \mathrm{d}x = \frac{3}{2}\sec^2 u\,\mathrm{d}u$

Change limits:

$x = 0 \Rightarrow u = \arctan(0 + 1) = \frac{\pi}{4}$

$x = \frac{\sqrt{3}-3}{2} \Rightarrow u = \arctan\left(\frac{2}{3}\left(\frac{\sqrt{3}-3}{2}\right) + 1\right) = \frac{\pi}{6}$

$\therefore I = \frac{1}{9}\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\frac{3}{2}\sec^2 u}{1+\tan^2 u}\,\mathrm{d}u$

$= \frac{1}{6}\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sec^2 u}{\sec^2 u}\,\mathrm{d}u$

$= \frac{1}{6}\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} 1\,\mathrm{d}u$

$= \frac{1}{6}[u]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$

$= \frac{1}{6}\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$

$= \frac{1}{6} \times \frac{\pi}{12}$

$= \frac{\pi}{72}$

7) d)

$I = \int_3^4 \frac{1}{-4x^2+28x-40}\,\mathrm{d}x$

$-4x^2 + 28x - 40$

$\equiv -4(x^2 - 7x) - 40$

$\equiv -4\left(\left(x - \frac{7}{2}\right)^2 - \frac{49}{4}\right) - 40$

$\equiv -4\left(x - \frac{7}{2}\right)^2 + 49 - 40$

$\equiv -4\left(x - \frac{7}{2}\right)^2 + 9$

$\therefore I = \int_3^4 \frac{1}{9-4\left(x-\frac{7}{2}\right)^2}\,\mathrm{d}x$

$= \int_3^4 \frac{1}{9\left(1-\frac{4}{9}\left(x-\frac{7}{2}\right)^2\right)}\,\mathrm{d}x$

$= \frac{1}{9}\int_3^4 \frac{1}{1-\left(\frac{2}{3}x-\frac{7}{3}\right)^2}\,\mathrm{d}x$

Use Integration by Substitution:

Let $\tanh u = \frac{2}{3}x - \frac{7}{3} \Rightarrow \operatorname{sech}^2 u \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{3} \Rightarrow \mathrm{d}x = \frac{3}{2}\operatorname{sech}^2 u\,\mathrm{d}u$

Change limits:

$x = 4 \Rightarrow u = \operatorname{artanh}\left(\frac{2}{3}\times 4 - \frac{7}{3}\right) = \operatorname{artanh}\frac{1}{3}$

$x = 3 \Rightarrow u = \operatorname{artanh}\left(\frac{2}{3}\times 3 - \frac{7}{3}\right) = \operatorname{artanh}\left(-\frac{1}{3}\right)$

$\therefore I = \frac{1}{9}\int_{\operatorname{artanh}\left(-\frac{1}{3}\right)}^{\operatorname{artanh}\frac{1}{3}} \frac{\frac{3}{2}\operatorname{sech}^2 u}{1-\tanh^2 u}\,\mathrm{d}u$

$= \frac{1}{6}\int_{\operatorname{artanh}\left(-\frac{1}{3}\right)}^{\operatorname{artanh}\frac{1}{3}} \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\,\mathrm{d}u$

$= \frac{1}{6}\int_{\operatorname{artanh}\left(-\frac{1}{3}\right)}^{\operatorname{artanh}\frac{1}{3}} 1\,\mathrm{d}u$

$= \frac{1}{6}[u]_{\operatorname{artanh}\left(-\frac{1}{3}\right)}^{\operatorname{artanh}\frac{1}{3}}$

$= \frac{1}{6}\left(\operatorname{artanh}\frac{1}{3} - \operatorname{artanh}\left(-\frac{1}{3}\right)\right)$

$= \frac{1}{6}\left(\frac{1}{2}\ln\left(\frac{1+\left(\frac{1}{3}\right)}{1-\left(\frac{1}{3}\right)}\right) - \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{3}\right)}{1-\left(-\frac{1}{3}\right)}\right)\right)$

$= \frac{1}{6}\left(\frac{1}{2}\ln 2 - \frac{1}{2}\ln\frac{1}{2}\right)$

$= \frac{1}{12}\left(\ln 2 - \ln\frac{1}{2}\right)$

$= \frac{1}{12}\ln\left(\frac{2}{\frac{1}{2}}\right)$

$= \frac{1}{12}\ln 4$

$= \frac{1}{12}\ln 2^2$

$= \frac{2}{12}\ln 2$

$= \frac{1}{6}\ln 2$

8)

$1 - x^4 \equiv (1 - x^2)(1 + x^2) \equiv (1 - x)(1 + x)(1 + x^2)$

$\frac{12x^2}{1-x^4} \equiv \frac{12x^2}{(1-x)(1+x)(1+x^2)} \equiv \frac{A}{1-x} + \frac{B}{1+x} + \frac{Cx+D}{1+x^2}$

$\Rightarrow 12x^2 \equiv A(1 + x)(1 + x^2) + B(1 - x)(1 + x^2) + (Cx + D)(1 - x)(1 + x)$

Let $x = 1$: $12 = 4A \Rightarrow A = 3$

Let $x = -1$: $12 = 4B \Rightarrow B = 3$

Let $x = 0$: $0 = 3 + 3 + D \Rightarrow D = -6$

Let $x = 2$: $48 = 45 - 15 - 3(2C - 6) \Rightarrow 2C - 6 = -6 \Rightarrow C = 0$

So $\frac{12x^2}{1-x^4} \equiv \frac{3}{1-x} + \frac{3}{1+x} - \frac{6}{1+x^2}$

$\therefore \int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{12x^2}{1-x^4}\,\mathrm{d}x = \int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \left(\frac{3}{1-x} + \frac{3}{1+x} - \frac{6}{1+x^2}\right)\mathrm{d}x$

$= \left[-3\ln|1 - x| + 3\ln|1 + x| - 6\arctan x\right]_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}}$

$= \left[3\ln\left|\frac{1+x}{1-x}\right| - 6\arctan x\right]_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}}$

$= \left(3\ln\left|\frac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}}\right| - 6\arctan\frac{1}{\sqrt{3}}\right) - \left(3\ln\left|\frac{1+\left(-\frac{1}{\sqrt{3}}\right)}{1-\left(-\frac{1}{\sqrt{3}}\right)}\right| - 6\arctan\left(-\frac{1}{\sqrt{3}}\right)\right)$

$= \left(3\ln\left(2 + \sqrt{3}\right) - \pi\right) - \left(3\ln\left(2 - \sqrt{3}\right) + \pi\right)$

$= 3\ln\left(\frac{2+\sqrt{3}}{2-\sqrt{3}}\right) - 2\pi$

$= 3\ln\left(7 + 4\sqrt{3}\right) - 2\pi$